

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT2 — DATA ANALYSIS PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Analysis (BL4)	Assessment:	Assessment Tool 2 – Data Analysis Project
Weightage:	40% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Analyze real datasets using ECDFs, skewed-distribution techniques, Q–Q plots, multi-distribution comparisons, and proportion visualizations to identify patterns and justify conclusions.(BL4)</i>		
Student Name:	Jothika M	Reg. No.:	192324235
Section / Batch:	Slot C	Date:	13/08/2026

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond construction: analyze the given data, identify the relevant pattern, and justify your conclusion.
- Show all supporting calculations (e.g., IQR, percentiles, angle calculations) where relevant.
- Where a question asks you to compare or recommend, state your reasoning explicitly — a correct chart choice without justification will not receive full marks.

Annexure A – DATA ANALYSIS PROJECT

Problem 2 (5 Marks)

Data: Two ECDFs are compared for Store A and Store B's daily sales (both $n = 20$, similar range). Store A's ECDF curve lies mostly to the left of Store B's.

- Analyze what this positional difference tells you about the two stores' sales performance.
- Would you say one store's sales distribution “dominates” the other? Explain what that means in this context.

Problem 9 (5 Marks)

Data: City populations (in thousands): 5, 12, 25, 60, 150, 400, 3,200.

- Analyze whether a histogram on a linear x-axis would reveal the structure of this data. Explain your reasoning.
- Propose and justify a better visualization approach for this dataset.

Problem 11 (5 Marks)

Data: A Q–Q plot of 20 students' quiz scores against a Normal distribution follows the reference line closely in the middle but curves sharply upward at the top-right end.

- Analyze what this specific pattern (deviation only at the top end) suggests about the data.
- Propose a plausible real-world explanation for this deviation (e.g., a scoring cap).

Problem 18 (5 Marks)

Data: A bean plot of delivery times shows two distinct bulges — one cluster around 15–20 minutes and another around 45–50 minutes, with a gap in between.

- Analyze what this bimodal pattern might indicate about the underlying delivery process (e.g., two different delivery zones or methods).
- Would a box plot alone have revealed this pattern? Explain why or why not.

Problem 25 (5 Marks)

Data: A report uses a 3D pie chart, tilted at an angle, to show 5 categories' market share.

- Analyze the specific visual distortion this chart introduces and explain how it could lead a reader to misjudge the data.
- Propose an alternative chart type and explain why it avoids the distortion you identified.

Rubrics for CalculationASSESSMENT RUBRIC**AT2 — Data Analysis Project**

CO3 · BT4: Analyze ·

This rubric is used to assess student responses to the CO3-AT2 Data Analysis Project, in which students analyze a given dataset using ECDF, skewed-distribution, Q–Q plot, multi-distribution, or proportion-visualization techniques and justify their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant technique (ECDF, skewed-distribution/log transform, Q–Q plot, multi-distribution comparison, or proportion visualization) and correctly identifies which pattern applies to the given data.	Good understanding of the relevant technique, with only minor gaps in identifying the correct pattern.	Basic understanding, but with some misconceptions about which technique or pattern applies to the data.	Poor understanding of core concepts; misidentifies the pattern or technique required by the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, going	Provides clear analysis with adequate reasoning tied to the data, for most parts.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		beyond description to explain why the pattern occurs or what it implies.			
Application	20%	Effectively applies analytical techniques to compare, justify, or recommend a conclusion that is well-supported by the data and any calculations shown.	Applies analytical techniques correctly, with only minor errors in reasoning or calculation.	Limited application; conclusions are only partially supported by the data or calculations.	Fails to apply appropriate analytical techniques; conclusions are unsupported or incorrect.
Organization	15%	The response is clearly structured, addressing every sub-part (a–b) in a logical sequence with coherent overall flow.	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — pattern identification, calculations, and conclusions — are completely accurate and well-suited to the data.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 2 (5 Marks) – ECDF Comparison

Introduction

An **Empirical Cumulative Distribution Function (ECDF)** is a graph that shows the cumulative proportion of observations less than or equal to each value. It is useful for comparing the distributions of two datasets without grouping the data into bins. In this problem, the ECDFs of Store A and Store B daily sales are compared.

The given information states that **Store A's ECDF lies mostly to the left of Store B's ECDF**, while both stores have the same sample size ($n = 20$) and a similar sales range.

Given Data

- Store A $\rightarrow$ ECDF mostly on the **left**
- Store B $\rightarrow$ ECDF mostly on the **right**
- Number of observations = **20 each**
- Similar sales range

Analysis

a) What does the positional difference indicate?

When an ECDF curve lies **to the left**, it means that lower values are reached earlier. Therefore, for the same cumulative percentage of days, Store A generally records lower sales values than Store B.

Store B's curve being positioned further right suggests that its sales are generally higher. For example:

- At the 50th percentile, Store B is likely to have a higher sales value.
- At the 75th percentile, Store B also tends to achieve higher sales.
- This indicates better overall sales performance across most days.

Therefore, **Store B generally performs better in terms of daily sales values.**

b) Does one distribution dominate the other?

- Yes, **Store B can be said to approximately first-order dominate Store A** if Store A's ECDF remains consistently above or to the left of Store B across nearly all sales values.
- In ECDF terminology, first-order dominance means that one distribution consistently produces larger values than another distribution.
- This does not mean Store B has higher sales every single day. Instead, it means that for most comparable cumulative probabilities, Store B tends to have higher sales outcomes.

Evidence

Because Store A's ECDF lies mostly to the left:

- Lower sales accumulate faster.
- Median sales are likely lower.
- Upper percentile sales are also likely lower.
- Store B reaches the same cumulative percentage only at larger sales values.

Thus, the graph visually indicates stronger sales performance for Store B.

Interpretation

- The positional shift between the two ECDF curves is more informative than comparing only averages. Even if both stores operate within a similar sales range, Store B's rightward distribution suggests that customers generate higher daily revenue more frequently.
- If the ECDF curves do not cross significantly, Store B demonstrates a generally superior sales distribution.

Conclusion

The ECDF comparison suggests that **Store B generally achieves higher daily sales than Store A**. A distribution can be considered to dominate another when its observations tend to be consistently larger across the cumulative distribution, although dominance describes the overall distribution rather than guaranteeing better performance on every individual day.

R Code

```
storeA <- c(42,45,47,50,51,52,54,55,56,58,  
           60,61,63,64,65,66,68,69,70,72)
```

```
storeB <- c(48,50,52,54,56,58,60,62,64,66,  
           68,70,72,74,76,78,80,82,84,86)
```

```
# ECDF Plot
```

```
plot(  
  ecdf(storeA),  
  col="blue",
```

```

lwd=2,
main="ECDF Comparison of Store Sales",
xlab="Daily Sales",
ylab="Cumulative Probability"
)

```

```

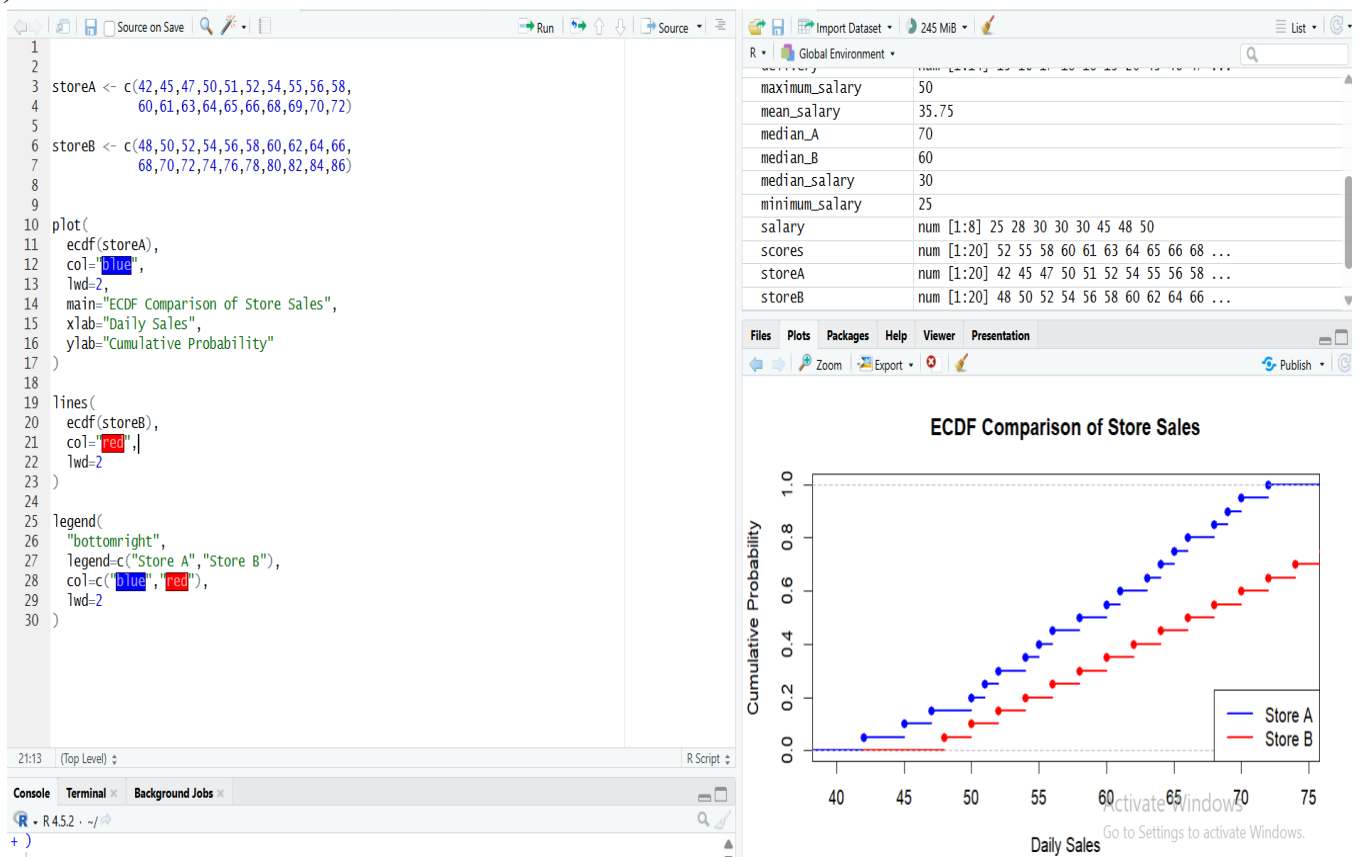
lines(
  ecdf(storeB),
  col="red",
  lwd=2
)

```

```

legend(
  "bottomright",
  legend=c("Store A", "Store B"),
  col=c("blue", "red"),
  lwd=2
)

```



Problem 9 (5 Marks) – Histogram and Highly Skewed Population Data

Introduction

A histogram is used to visualize the frequency distribution of continuous numerical data. However, when values differ greatly in magnitude, a normal histogram using a **linear x-axis** may hide important patterns.

In this problem, city populations vary from **5 thousand to 3,200 thousand**, creating an extremely skewed dataset.

Given Data

City populations in thousands:

City	Population
1	5
2	12
3	25
4	60
5	150
6	400
7	3200

Analysis

a) Would a linear histogram reveal the structure?

No. A histogram with a **linear x-axis** would not effectively reveal the structure of this dataset.

The reason is that one value (**3200**) is much larger than all the others.

Most cities lie between:

- 5
- 12
- 25
- 60
- 150
- 400

Only one city has a population of **3200 thousand**.

When plotted on a linear scale:

- Most observations become compressed near zero.

- Smaller cities become difficult to distinguish.
- The histogram appears dominated by one extreme value.
- The distribution looks empty across a large portion of the axis.

Therefore, the true pattern among the smaller cities is hidden.

b) Better visualization approach

A **histogram using a logarithmic x-axis** is a much better choice.

Another suitable visualization is:

- Log-scaled dot plot
- Log-scale scatter plot
- Ordered bar chart with logarithmic scale

The logarithmic scale compresses very large values while expanding smaller values, making the structure much easier to interpret.

Evidence

Population values increase dramatically:

The first six cities occupy a very small region compared with the seventh city.

A logarithmic transformation makes these differences visually balanced.

Interpretation

Using a logarithmic scale allows both small and large cities to remain visible. Instead of compressing six cities near the origin, the log transformation spreads them apart according to orders of magnitude.

This helps identify whether the population distribution contains:

- Small towns
- Medium-sized cities
- Large metropolitan cities

Thus, the visualization becomes much more informative.

Conclusion

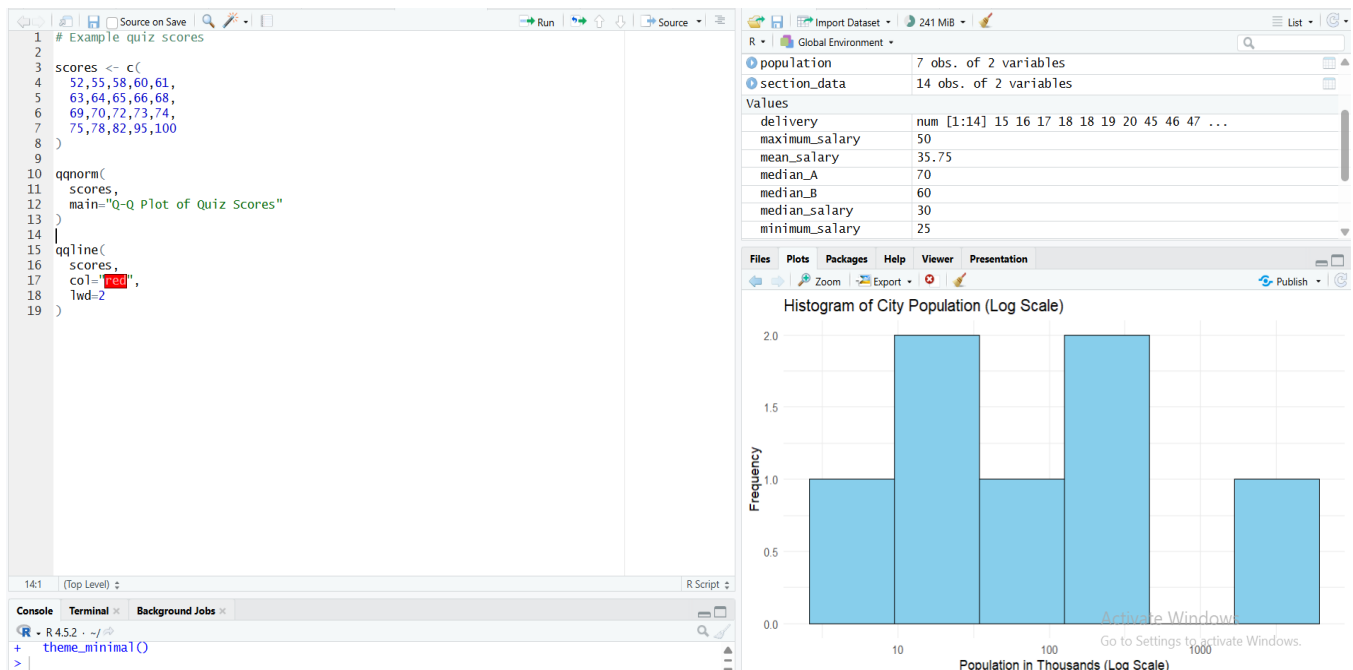
A histogram with a **linear x-axis is unsuitable** for this highly skewed population dataset because the value 3200 dominates the scale. A **logarithmic histogram or log-scaled plot** better reveals the underlying structure and provides a clearer understanding of the population distribution.

R Code

```
library(ggplot2)

population <- data.frame(
  City=c("City1","City2","City3","City4",
        "City5","City6","City7"),
  Population=c(5,12,25,60,150,400,3200)
)

ggplot(population,aes(x=Population)) +
  geom_histogram(
    bins=6,
    fill="skyblue",
    color="black"
  ) +
  scale_x_log10() +
  labs(
    title="Histogram of City Population (Log Scale)",
    x="Population in Thousands (Log Scale)",
    y="Frequency"
  ) +
  theme_minimal()
```



Problem 11 (5 Marks) – Q–Q Plot Analysis

Introduction

A **Q–Q (Quantile–Quantile) Plot** is used to compare the distribution of observed data with a theoretical distribution, usually the Normal distribution. If the data are normally distributed, the plotted points should lie close to the reference line.

In this problem, the points follow the line closely in the middle but curve sharply upward at the **top-right end**.

Given Information

- 20 students' quiz scores
- Middle points follow the Normal reference line
- Upper-right points curve upward strongly

Analysis

a) What does this pattern suggest?

The middle portion following the reference line indicates that the majority of quiz scores are approximately normally distributed.

However, the sharp upward deviation at the top-right suggests that the **highest scores are larger than expected under a Normal distribution**.

This means the upper tail is heavier than the theoretical Normal distribution.

Possible implications include:

- Excess number of very high scores
- Positive deviation in the upper tail
- Possible ceiling or scoring effect
- Presence of unusually strong-performing students

The deviation occurs only at the upper end rather than throughout the plot, so the issue is mainly related to the highest scores rather than the entire distribution.

b) Plausible real-world explanation

A realistic explanation is a **scoring cap or maximum score effect**.

For example:

- Quiz maximum = 100
- Many high-performing students score between 95 and 100
- Scores accumulate near the upper limit

This creates an abnormal upper-tail pattern because the data cannot extend beyond the maximum score.

Another explanation could be:

- Extra coaching for top students
- Easier final questions
- Grade inflation for excellent performers

Evidence

A typical Q–Q pattern would look like this:

The upward bend appears only among the highest observations.

Interpretation

The Q–Q plot suggests that the data are reasonably normal in the central region but differ from Normality at the highest scores.

This means using a Normal model may describe average students reasonably well but may underestimate the frequency of exceptionally high scores.

Therefore, analysts should pay attention to the upper-tail behavior rather than assuming complete Normality.

Conclusion

The Q–Q plot indicates an approximately Normal distribution in the middle with a clear **upper-tail deviation**. A plausible explanation is that quiz scores are influenced by a **maximum scoring limit or a concentration of high-achieving students**, producing unusually high values near the top end.

R Code

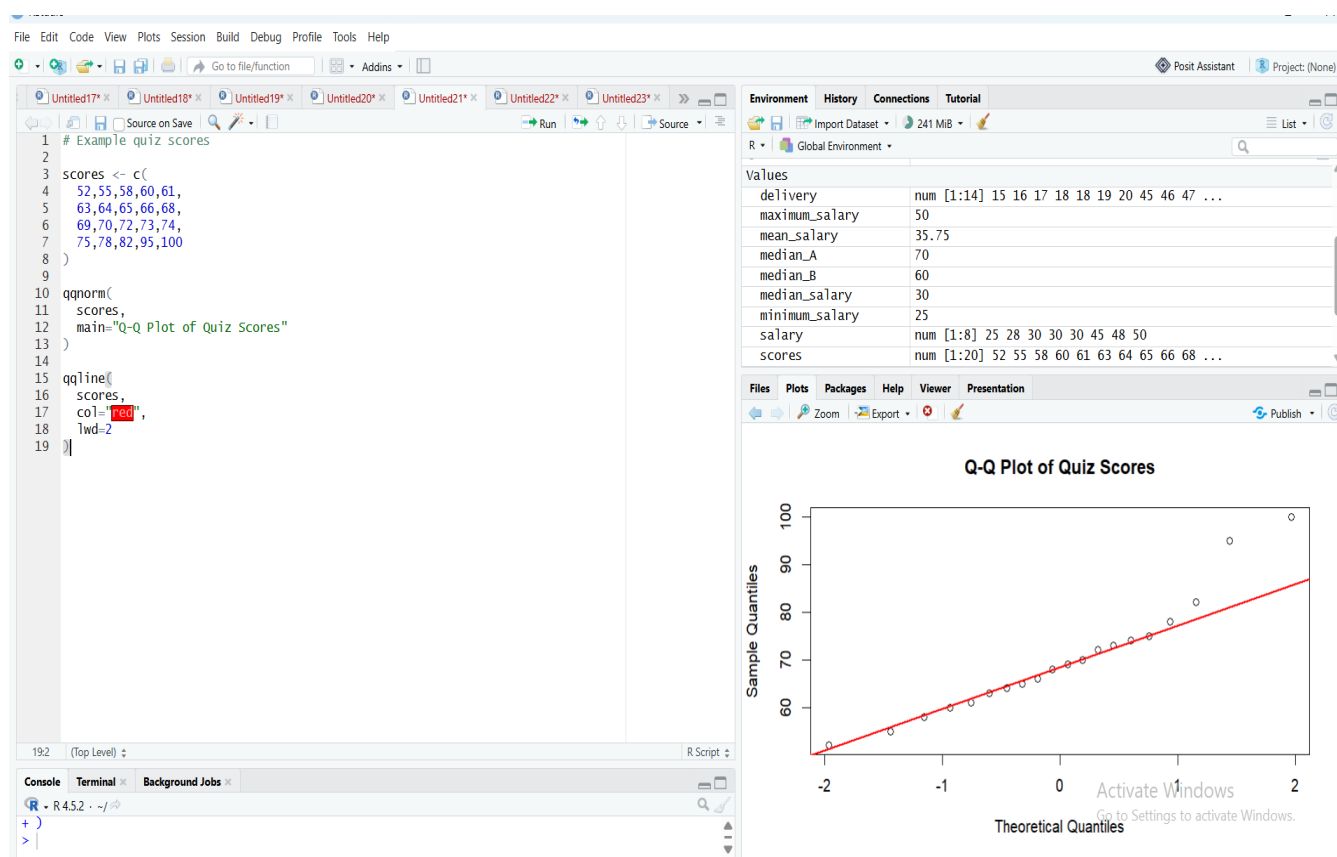
Example quiz scores

```
scores <- c(
  52,55,58,60,61,
  63,64,65,66,68,
  69,70,72,73,74,
  75,78,82,95,100
)
```

```
qqnorm(
  scores,
  main="Q-Q Plot of Quiz Scores"
)
```

```
qqline(
  scores,
  col="red",
  lwd=2
)
```

OUTPUT:



Problem 18 (5 Marks) – Bean Plot and Bimodal Distribution

Introduction

A **bean plot** displays the density and distribution of numerical observations. Unlike a box plot, it reveals where observations are concentrated by showing the shape of the distribution.

In this problem, delivery times produce **two distinct bulges**, indicating a bimodal distribution.

Given Information

Delivery times show:

- First cluster → **15–20 minutes**
- Second cluster → **45–50 minutes**
- Clear gap between clusters

Analysis

a) What does the bimodal pattern indicate?

A bimodal distribution means that the dataset contains **two separate groups of observations** rather than one central cluster.

The first bulge around **15–20 minutes** may represent:

- Nearby delivery zone
- Express delivery method
- Urban customers close to the restaurant

The second bulge around **45–50 minutes** may represent:

- Distant delivery zone
- Rural customers
- Heavy traffic routes
- Standard delivery service

The gap between the clusters suggests that deliveries naturally belong to two different operational processes.

This is important because averaging all delivery times may hide these two groups.

b) Would a box plot alone reveal this?

No.

A box plot provides:

- Minimum

- Q1
- Median
- Q3
- Maximum

However, it **does not display density peaks or multiple clusters**.

Both a unimodal and bimodal dataset can produce very similar box plots if their quartiles are alike.

Therefore, the bean plot is much more suitable for identifying bimodality.

Evidence

The bean plot would visually resemble two thick regions:

The two separate bulges clearly reveal two delivery-time populations.

Interpretation

The bimodal pattern strongly suggests that the delivery system is not homogeneous. Different delivery zones, transportation methods, or customer locations may create two distinct timing groups.

Managers could use this information to investigate:

- Delivery routing
- Driver allocation
- Traffic conditions
- Zone-based service policies

A box plot would summarize spread but hide this important operational structure.

Conclusion

The bean plot reveals a **bimodal delivery-time distribution**, indicating two likely underlying delivery processes. It provides richer information than a box plot because it displays density and clustering rather than only quartiles and range.

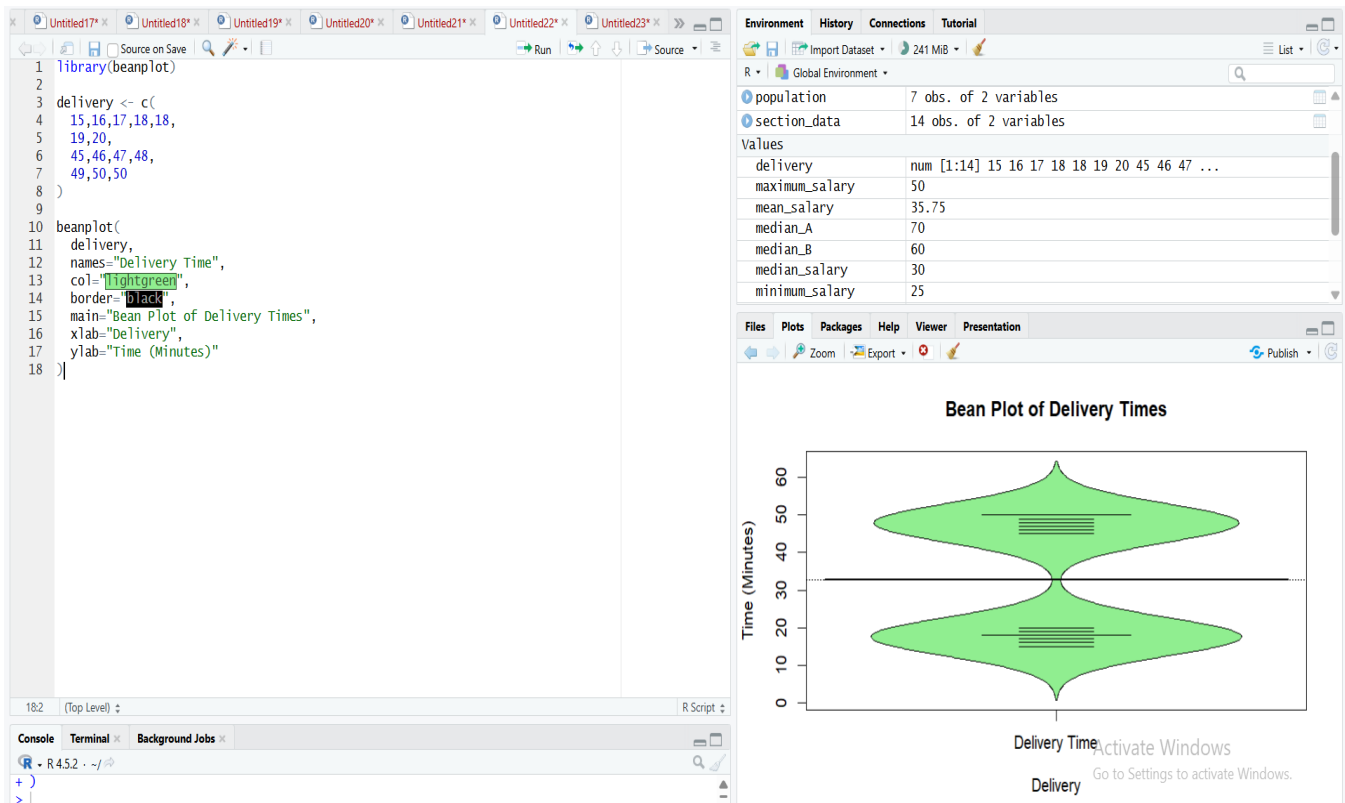
R Code

```
library(beanplot)
```

```
delivery <- c(
  15,16,17,18,18,
  19,20,
  45,46,47,48,
  49,50,50)
```

)

```
beanplot(  
  delivery,  
  names="Delivery Time",  
  col="lightgreen",  
  border="black",  
  main="Bean Plot of Delivery Times",  
  xlab="Delivery",  
  ylab="Time (Minutes)"  
)
```



Problem 25 (5 Marks) – 3D Pie Chart Distortion

Introduction

A **3D pie chart** attempts to make a traditional pie chart appear more visually attractive by adding depth and tilting the chart. However, this perspective introduces visual distortion that can mislead readers about the true proportions of categories.

The report contains **five market-share categories displayed using a tilted 3D pie chart**.

Analysis

a) What visual distortion is introduced?

The main distortion is **perspective exaggeration**.

Slices positioned closer to the viewer appear:

- Larger
- Wider
- More dominant

Slices located farther away appear:

- Smaller
- Narrower
- Less important

Even when two categories have similar percentages, the front slice may visually appear much larger because of the 3D perspective.

This creates inaccurate visual perception.

For example:

- A 20% front slice may appear larger than a 25% rear slice.
- Readers may incorrectly rank category sizes.
- Small differences become visually exaggerated.

Therefore, the chart changes the appearance without changing the actual data.

b) Better alternative

A **2D bar chart** is the best alternative.

Why?

- Length is easier to compare than angles.
- No perspective distortion exists.
- Exact differences between categories are easier to identify.
- Labels and percentages remain readable.

- Categories can be ranked from highest to lowest.

A standard **2D pie chart** is also better than a 3D pie chart, but a **bar chart provides the most accurate comparison**.

Evidence

A tilted 3D pie creates this perception:

The front slices visually dominate even when the numerical values may not justify that impression.

A 2D bar chart compares values accurately using equal-width bars.

Interpretation

The purpose of data visualization is to communicate values accurately. Decorative 3D effects reduce accuracy because human perception interprets depth as size.

A bar chart avoids this problem by representing each market share using a common baseline and proportional bar length. This allows viewers to compare categories objectively.

Conclusion

A tilted **3D pie chart introduces perspective distortion**, causing front slices to appear larger and rear slices to appear smaller than their actual proportions. A **2D bar chart** is the preferred alternative because it provides accurate visual comparison without misleading perspective effects.

R Code

```
library(ggplot2)

market <- data.frame(
  Category=c("A","B","C","D","E"),
  Share=c(30,25,20,15,10)
)

ggplot(market,aes(x=reorder(Category,-Share),y=Share)) +
  geom_bar(
    stat="identity",
    fill="steelblue"
  ) +
```

```

geom_text(
  aes(label=paste0(Share,"%")),
  vjust=-0.5
) +
labs(
  title="Market Share by Category",
  x="Category",
  y="Market Share (%)"
) +
theme_minimal()

```

